CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY
CHANGA

B.Tech. EE Sem - IV Examination, 2010-11
EE - 204 Electrical Power Systems-I

Date: 07/12/2011, Wednesday **Time: 10 to 1 p.m** **Maximum Marks: 70**

Instructions:

1. Attempt all questions from both sections.
2. Use separate answer book for each sections.
3. Figures to the right indicate **full** marks.
4. Assume suitable additional data, if necessary

SECTION – I

Q-1 What is electric power supply scheme? Draw & explain single line diagram of a typical a.c power supply scheme & Discuss advantage of a.c. transmission. **7**

Q-2 A. Define following terms: **5**

(i) Connected load
(ii) Maximum demand
(iii)Demand factor
(iv)Load factor
(v) Diversity factor

B. Discuss the importance of high load factor. **3**

C. A generating station has a maximum demand of 50,000 KW. Calculate the cost per unit generated from the following data: **4**
Capital cost = Rs 95 X 10^6 ; Annual cost of fuel, oil etc. = Rs 9 X 10^6
Annual load factor=40%; ; Taxes, wages and salaries etc. = Rs 7.5 X 10^6
Annual interest & depreciation=12% .

D. What is meant by efficiency of an string? And list out the methods, that are used for improving string efficiency. **2**

OR

Q-2 A. Derive the equation of the optimum power factor. **5**

B. List out the advantages and dis-advantages of static capacitor as a power factor improvement device. **2**

C. The load on an installation is 800 KW, 0.8 lagging which works for 3000 hours per annum. The tariff is Rs. 100 per KVA plus 20 paisa per KWh. If the power factor is improved to 0.9 lagging by means of loss-free capacitors costing Rs. 60 per KVAR, calculate the annual saving effected. Allow 10% per annum for interest and depreciation on capacitor. **4**

D. A 3-phase overhead transmission line is being supported by three disc insulators. The potential across top unit (i.e. near to the tower) and middle unit are 8 KV and 11 KV respectively. Calculate (i) the ratio of capacitance between pin and earth to the self-capacitance of each unit (ii) the line voltage and (iii) string efficiency. **3**

(P. T. O)

Q-3 **A.** Discuss the advantage of interconnected grid system. 3

B. Describe the properties of line supports. List out the different types of line support with their peculiarity and area of application. 6

C. Explain the limitation of pin type insulator also list out the advantages of suspension type insulators. 5

OR

Q-3 **A.** List out the requirements of the cable. Also draw well labelled cross section of three core cable and explain function of each part and material used for that part. 7

B. Three single core cables are use for 3-phase 11 KV, 50 Hz system has conductor area of 0.645 cm^2 and the internal diameter of sheath is 2.18 cm. The permittivity of the dielectric used in the cable is 3.5. Find the maximum and minimum electrostatic stresses in the cable. Also find the capacitance and charging current per phase of the above cable per km length. 7

SECTION – II

Q-4 A 3-phase, 50 Hz, 150 km line has a resistance, inductive reactance and capacitive shunt admittance of 0.1 Ω, 0.5 Ω and $3X10^{-6}$ □ per km per phase. If the line delivers 50 MW at 110 KV and 0.8 p.f. lagging, determine the sending end voltage and current. Assume a nominal π circuit for the line. 7

Q-5 **A.** A 33 KV single circuit, 3-phase transmission line has the ABCD parameters A= D= $1 \angle 0°$, B=$11.18 \angle 63.43°$ Ω. The line is to deliver 7.5 MVA at 0.85 pf lagging at the load end. The receiving end voltage is 32 KV (line to line). How much active and reactive power is to be despatched from the sending end? 7

B. Draw & explain nominal T model and vector diagram of a medium transmission line. 4

C. Explain ferranti effect with its phasor diagram 3

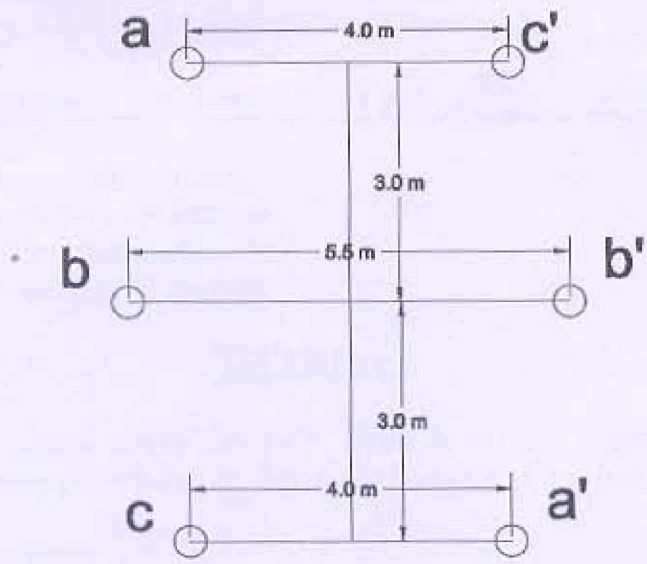
OR

Q-5 **A.** A 200 km, 3-phase, 50 Hz transmission line has the following data: 7
A=D= $0.938\angle 1.2°$, B=$131.2\angle 72.3°$ Ω/phase; C=$0.001\angle 90°$ Siemens/phase. The sending end voltage is 230 KV. Determine (i) the receiving end voltage when the load is disconnected, (ii) the line charging current (iii) the maximum power that can be transmitted at a receiving end voltage of 220 KV and the corresponding load reactive power required at the receiving end.

B. Draw & explain π model and vector diagram of a medium transmission line. 5

C. Define: Short and medium transmission line 2

Q-6 A. Find the inductance per phase per km of double circuit 3-phase line shown in figure. The conductors are transposed and are of radius 0.75 cm each. The phase sequence is ABC. 7



B. Calculate the inductance of each conductor in a 3-phase, 3-wire system when the conductors are arranged in a horizontal plane with spacing such that D_{31}=4 m; D_{12}=D_{23}=2 m. The conductors are transposed and have a diameter of 2.5 cm. 5

C. Why is transmission lines transposed? 2

OR

Q-6 A. Write technical short note on: Bundled conductor. 7

B. Write technical short note on: Effect of earth on the capacitance of a transmission line 7